BimmerStein ECU Tool

BMW MS41 Programming, Diagnostics, and Recovery

USER MANUAL

Version 0.1.0b14 - Windows x64

OFF-ROAD USE ONLY

Read the safety section before connecting to an ECU. Use stable power, confirm the selected operation, and never interrupt an active flash write.

Safety first

Read this section before connecting to an ECU or opening a write workflow.

IMPORTANT: OFF-ROAD, COMPETITION, RESEARCH, AND BENCH USE ONLY. Do not use this software to modify a vehicle operated on public roads. The user is responsible for compliance with applicable emissions, safety, registration, and other laws.

- Use a stable, regulated vehicle or bench power supply.
- Confirm the selected COM port, wiring mode, ECU variant, flash-chip family, and file size.
- Keep ignition ON and maintain power throughout an active write.
- Do not disconnect the FTDI adapter while a flash operation is running.
- A backup is recommended, but the Flash tab respects the operator's backup selection.
- Enable read-back verification when an independent byte comparison is required.
- Keep hardware BSL recovery available when testing boot-region or experimental firmware changes.

DANGER: If a write fails after erase has started, do not turn ignition off. Do not disconnect the adapter and do not close the application. Use **Retry Flash Recovery** while the retained high-speed or RAM-agent session is still active.

WARNING: Hardware BSL and armed Soft-BSL boot-region writes can rewrite code required for normal ECU startup. A failed boot-region write can leave the ECU unbootable until hardware BSL recovery.

After a successful flash

Every successful Flash-tab write ends with the same operator instruction:

1. Turn ignition OFF.
2. Wait at least 10 seconds.
3. Turn ignition ON.

The ignition cycle is an operational handoff, not a substitute for flash finalization or optional read-back verification.

Supported scope

Area	Supported scope	Important boundary
ECU families	BMW MS41.0, MS41.1, MS41.2, and MS41.3	Full-ROM conversion is supported across all four variants.
Normal communications	BMW DS2 over K-Line, 9600 baud, 8E2	DS2 is not KWP2000.
Stock fast transfers	Native-fast DS2 through an FTDI D2XX connection	The ECU-exact requested high rate is 187,500 baud.
Soft-BSL	Persistent loader plus RAM agents	Installation modifies firmware and requires the guided workflow.
Hardware BSL	Intel 28F200 and AMD/JEDEC 29F200/29F400	Uses a separate direct ASC0 tap, not the normal K-Line connection.
File sizes	256 KB full ROM and 24 KB tune	Choose the operation matching the file and intended region.
Host platform	Windows x64 installer or portable release	PyInstaller and Nuitka packages contain the same application features.

Flash-chip families

- **Intel 28F200** uses the Intel command set. Hardware-BSL erase/program operations require the correct external 12 V VPP/RP# supply.
- **AMD/JEDEC 29F200 and 29F400** use the AMD command set and are single-supply devices. Do not apply Intel VPP requirements to them.
- A 29F400 operation must also select the correct visible upper or lower half.

Install and start

Windows installer

1. Download either versioned Windows x64 installer.
2. Run it for a per-user installation with no administrator access required.
3. Install the driver for the intended FTDI adapter, then launch **BimmerStein ECU Tool**.

Assets whose names contain `-Nuitka` use the Nuitka backend. That installer uses a distinct product identity and installation directory so it can coexist with the PyInstaller build. Report the selected backend when describing startup or packaging behavior.

Portable Windows package

1. Verify the release ZIP checksum against its matching `.zip.sha256` file when supplied.
2. Extract the complete ZIP into a writable folder.
3. Run `BimmerStein ECU Tool.exe` from the extracted application folder.

Do not move only the executable. PyQt, protocol resources, patch descriptors, and RAM-agent payloads are stored under `_internal` in the PyInstaller package and beside the executable in the flat Nuitka package. Keep the selected package's complete extracted folder together.

The executable may not be code-signed. Windows can show an unknown-publisher warning. Confirm the release filename and matching `.zip.sha256` value before continuing.

Interface driver

Install the driver for the selected FTDI adapter. D2XX is preferred for stock native-fast DS2, Soft-BSL, and hardware BSL. When D2XX is unavailable, normal DS2 and supported hardware-BSL paths can fall back to pyserial at their compatible rates.

Data folders

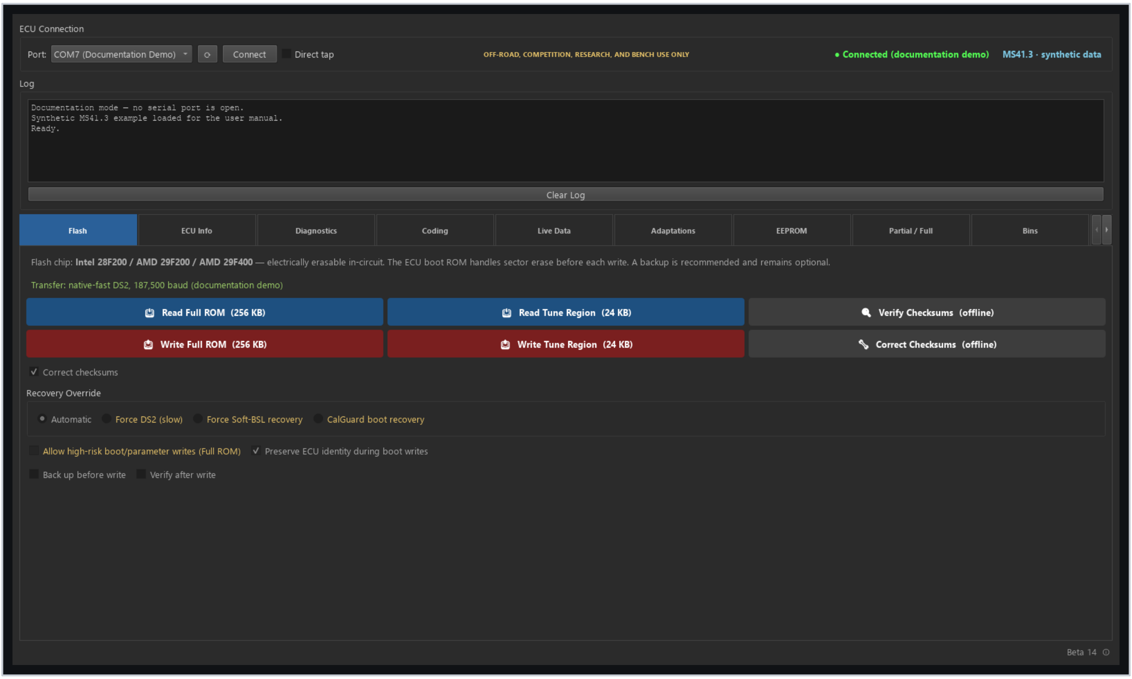
The portable application creates mutable data beside the executable:

- `backups/` for saved reads, generated images, and recovery material.
- `logs/` for session logs and diagnostic detail.

Every package includes `BimmerStein MS41 Patch Definitions.xml` beside the executable for use with RomRaider or BimmerStein Tuning Suite. The ROM Analyzer stores user-imported definitions under `%LOCALAPPDATA%\BimmerStein ECU Tool\definitions\`. This keeps the selected definition available when the portable application folder is replaced during an update.

Treat full ROMs and logs as private. They can contain a VIN, calibration identity, ECU identity, and operation history.

Main window



Main Flash workspace while disconnected

The connection bar remains visible above all tabs. It contains the normal DS2 COM selection, Connect button, direct-tap choice, connection state, ECU variant, and transfer-mode information. The shared log and progress controls remain visible below it.

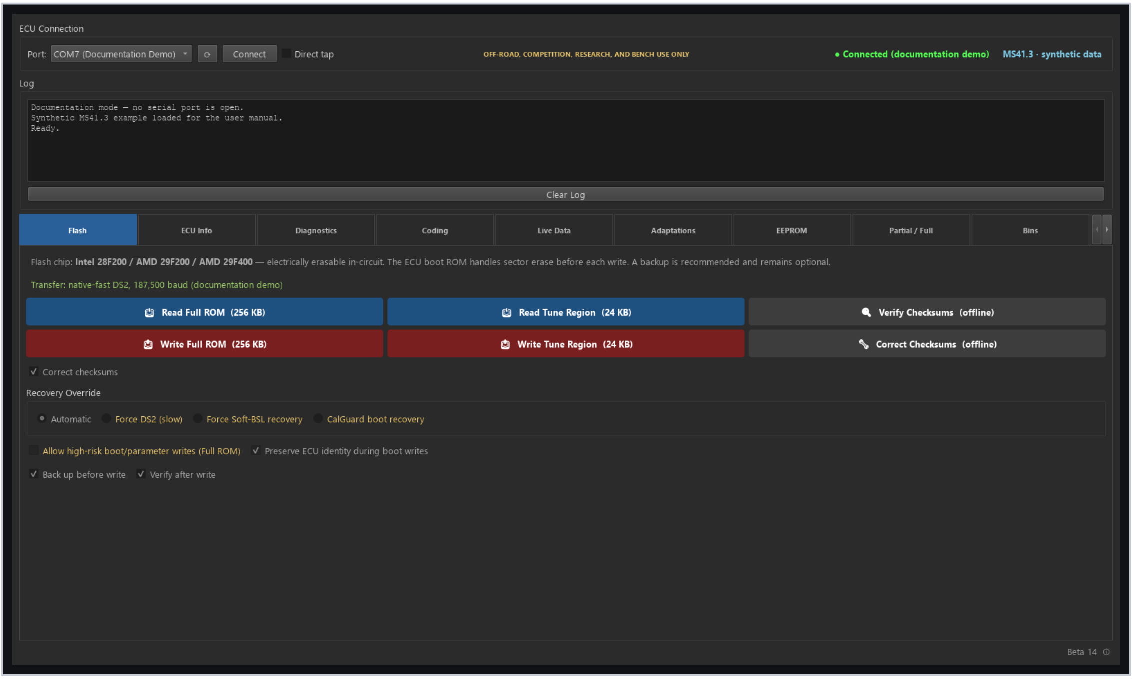
Normal K-Line versus direct tap

- Leave **Direct tap** clear for a normal single-wire K-Line or OBD-II adapter.
- Select it only for a full-duplex ASC0 connection that does not echo transmitted bytes.
- Choose the wiring mode before connecting. It is locked while a session owns the port.
- Hardware BSL has its own COM selector because it normally uses a separate adapter and wiring.

Tab order

Tab	Primary purpose
Flash	Full and tune reads/writes through the automatically selected transfer path.
ECU Info	Live identity, firmware, calibration, VIN, chip, and loader information.
Diagnostics	Scan verified vehicle-module addresses and read, export, or clear supported fault memories.
Coding	Read and change recognized module settings, or run the guided transmission conversion.
Live Data	Poll selected MS41 values and record CSV data.
Adaptations	Read fuel, throttle, and knock adaptations or reset selected learned values.
EEPROM	Inspect an exact 512-byte image and use the guarded ECU Agent or CH341A workflows.
Partial / Full	Convert between 24 KB tune and 256 KB full images.
Bins	Catalog reads, backups, generated images, and notes.
Patches	Compose, detect, migrate, or remove supported firmware patches.
ECU Config	Inspect and modify supported calibration configuration switches.
VIN / EWS	Separate VIN editing and EWS2 alignment workflows.
ROM Analyzer	Inspect a BIN offline without connecting to an ECU.
Soft-BSL	Install and operate the persistent Soft-BSL loader.
BSL-Unbricker	Hardware bootstrap reads, erase, program, and recovery.

Flash tab



Flash tab controls and transfer status

The Flash tab is the normal starting point for ECU reads and writes.

Choose the operation

- **Read Full** saves a single-pass 256 KB full ROM.
- **Read Tune** saves the 24 KB calibration region.
- **Write Full** programs the supported full-image regions while respecting boot-write controls.
- **Write Tune** programs the calibration region.

Automatic transfer selection

The application selects one of these routes:

1. **Soft-BSL** when the persistent loader is detected. It starts at the highest supported tier and retries lower Soft-BSL tiers only before erase.
2. **Native-fast DS2** on a compatible stock ECU through D2XX. A failed pre-erase high-rate check can restart the complete operation over normal DS2 only after the normal ECU state is confirmed.
3. **Normal DS2 at 9600** when neither accelerated route is available.

Fallback is allowed only before erase. Once erase may have started, the application retains the active session for recovery instead of changing transports.

Write options

- **Correct checksums** is enabled by default. MS41.3 boot and calibration checksums are corrected; its program checksum remains unchanged because stock program verification is disabled.
- **Back up before write** is optional and follows the operator's selection.
- **Verify after write** controls host-side byte-for-byte read-back verification.
- ECU-side flash finalization is independent of the optional host Verify checkbox.
- Boot/parameter writes remain separately armed because they carry a larger recovery risk.

Failure and retained-session recovery

Failure before erase

Before erase, Soft-BSL can retry the operation at a lower Soft-BSL baud tier. Native-fast DS2 can restart through normal DS2 at 9600 only after the normal low-rate ECU state has been confirmed.

Failure after erase

After erase begins, changing baud rate, reopening the port, or cycling ignition can discard the only live recovery path. The application therefore retains:

- The D2XX native-fast session for a native DS2 write failure.
- The loaded RAM agent and port ownership for a Soft-BSL write failure.
- The exact prepared target bytes required for a same-session retry.

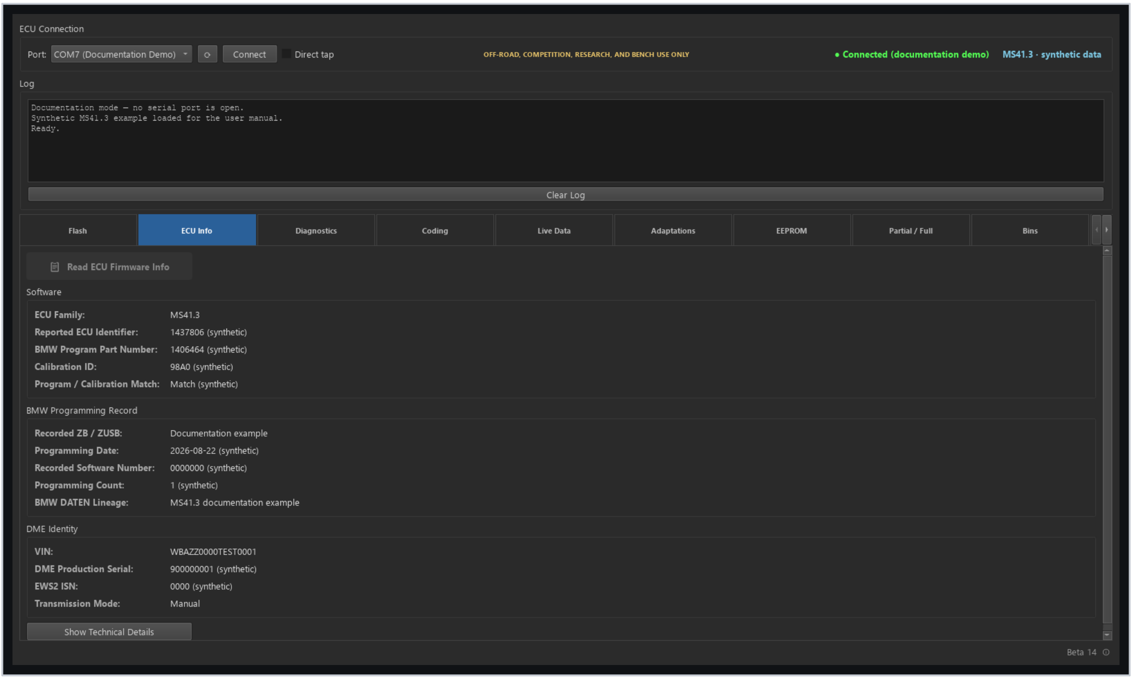
Follow the red recovery message exactly. Keep ignition ON and select **Retry Flash Recovery**.

When hardware BSL is required

Use hardware BSL when normal DS2 no longer starts and no retained Soft-BSL/native session is available. Hardware BSL runs from CPU bootstrap ROM and does not depend on valid flash contents.

> **DANGER:** Do not experiment with reset lines, ignition cycles, VPP, or chip-family selections during a retained recovery session. Preserve the session first; use hardware BSL only when the retained path is no longer available or cannot recover the target.

ECU Info, Diagnostics, and Live Data



Synthetic ECU information workspace

ECU Info

Use **Read ECU Information** after connecting. The tab can report the detected ECU variant, firmware/ECU ID, calibration ID, VIN, identity strings, flash-driver signature, flash-chip family, and Soft-BSL marker when available.

If MS41.2 and MS41.3 identification is ambiguous, use a full image or the stronger program and calibration markers rather than relying only on the shared ECU ID.

Diagnostics

1. Leave **Direct tap** clear and connect through normal K-Line.
2. Select **Scan Vehicle** for a read-only identification scan of the six verified module addresses. Cabin OBD may expose only powertrain modules; use the BMW 20-pin diagnostic path when applicable.
3. Select the responding module. For ABS/traction-control or automatic-transmission fault memory, select the exact fitted system profile. If no embedded profile matches, fault reading and clearing remain unavailable rather than using an assumed record layout.
4. Select **Read Faults**, then review the code, reference, system, status, description, and selected-fault detail.
5. Select **Export to Text** when a service record is required.
6. Record the faults and correct their cause before selecting **Clear Faults**. Read the module again after a non-engine clear request to confirm the result.

Live Data

Review the fixed parameter rows for plausible values and units. Fast telegram mode requests the mapped addresses in one batch; compatible mode reads the same mapped values in smaller blocks. CSV logging is written automatically under the application `logs/` directory while logging is enabled.

Fast telegram mode uses all 24 ECU slots. In addition to the core engine values, it reports EVAP purge duty, front-O2 and MAF input voltages, and operating states such as closed throttle, part/full load, deceleration fuel cut, and engine start. On MS41.3, the tool checks the runtime wideband flag once when polling starts. If enabled, the profile automatically reports actual AFR, target AFR, and the configured wideband input voltage; the table also identifies the selected input and whether narrowband emulation is active.

Coding and guided transmission conversion

Vehicle module coding

WARNING: HIGHLY EXPERIMENTAL — NOT VEHICLE TESTED. The Coding tab can change configuration in multiple vehicle modules. Built-in profiles and read-back checks reduce mistakes, but they do not prove a change is safe for a particular vehicle. Back up first, use stable power, keep the engine off, change only settings you understand, and be prepared to restore the original coding.

Module coding requires normal K-Line mode; Direct Tap reaches only the Engine ECU.

1. Choose the vehicle chassis and module, then select **Read Settings**.
2. The application identifies the connected coding version and exposes only an exact built-in profile. An unknown module version remains unavailable.
3. Change only understood settings. **Show advanced options** reveals reviewed equipment-specific settings and references; it does not enable raw-byte editing.
4. Select **Write Changes** and review the exact old-to-new values and battery-voltage notice.
5. Keep ignition ON and the engine OFF while writing. A successful operation reads the module back and requires the decoded settings to match. Cycle ignition before testing the change.

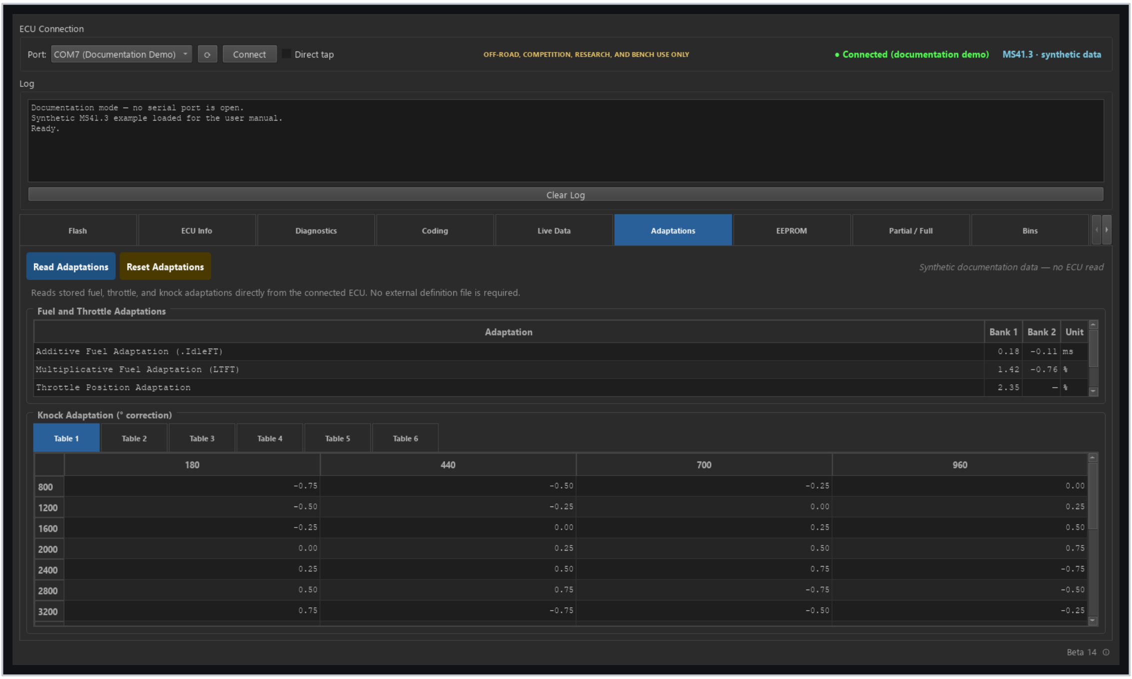
Guided transmission conversion

The compatibility check is read-only. It admits only an exact built-in chassis, engine-computer, vehicle-order, fitted-module, and transmission-computer combination; unresolved or conflicting state stops before a write.

1. Connect through normal K-Line, choose **Manual transmission** or **Automatic transmission**, and select **Check Compatibility**.
2. Review every reported owner, warning, blocking reason, and planned change. Complete the mechanical transmission and wiring work before continuing.
3. Select **I confirm the physical transmission swap is complete and the vehicle is safely parked**, then select **Convert to Manual** or **Convert to Automatic**.
4. Keep ignition ON and the engine OFF until the application explicitly requests a cycle. Before the first module write, the application durably archives the admitted before-state and starts a recovery journal. Each written owner is read back before the workflow advances.
5. When instructed, turn ignition OFF, wait at least 10 seconds, then turn ignition ON with the engine stopped.
6. Confirm the completed cycle and select **Verify Conversion**. Final verification independently reads every changed module and the applicable DME transmission record.

If a conversion is interrupted, do not start another plan. Use **Recover Interrupted Conversion...** to load the checked local record and either finish the reviewed target or restore the archived original coding. If the DME transmission-record result is unresolved, keep ignition ON and use the guided recovery; no success or rollback is assumed.

Adaptations

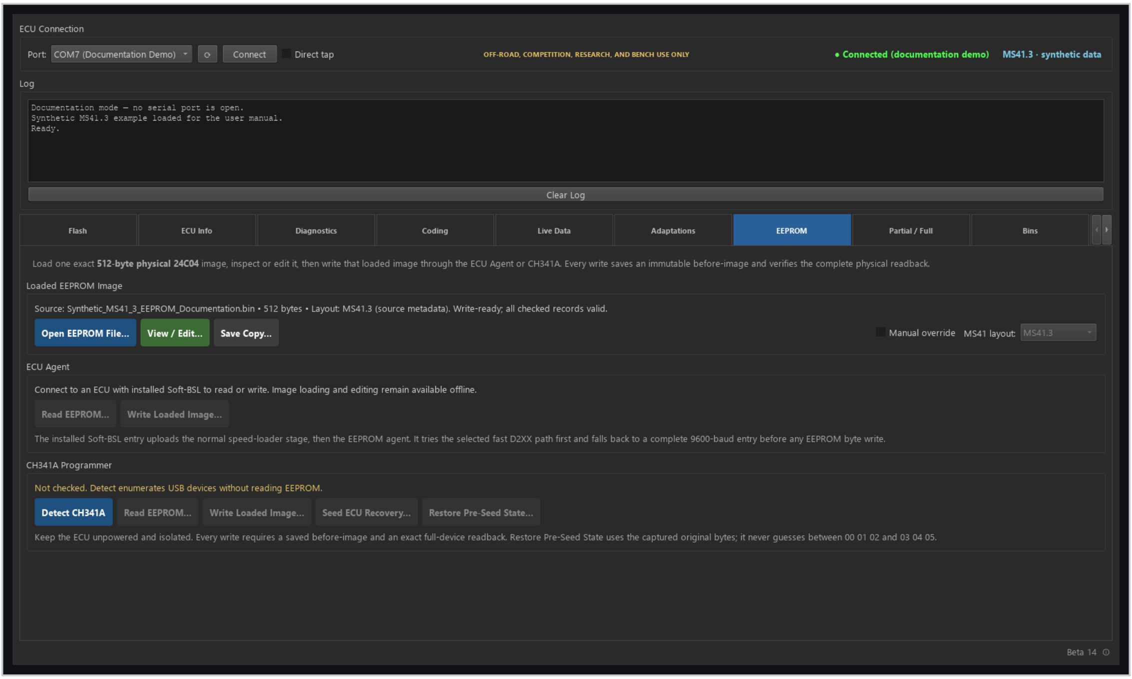


Adaptations tab with synthetic values

Read Adaptations reads stored fuel, throttle, and six 16-by-4 knock-adaptation tables from the connected Engine ECU. The exact ECU ID must have packaged address and axis mappings; the operation stops when those mappings are unavailable instead of estimating them. Fuel values without an exact mapping remain shown as –. The knock tables use load columns, RPM rows, and degrees of correction.

Reset Adaptations changes learned ECU state and requires confirmation. MS41.0 offers **All adaptations** only. Other supported families offer **All adaptations**, **Idle adaptation**, **Knock adaptation**, **Lambda / fuel trim adaptation**, or **Throttle adaptation**. The ECU relearns the selected values during subsequent operation.

EEPROM



Synthetic offline EEPROM workspace

The EEPROM tab works with one displayed 512-byte physical 24C04 image. Loading a file or an `EEPROM` entry from Bins replaces that same target; neither action writes to hardware.

> **DANGER:** Do not write a short live RAM mirror padded to 512 bytes, a uniform failed capture, or an image whose exact MS41 layout is unknown. EEPROM data can contain vehicle identity and coding; keep captures private.

Load and inspect an image

1. Select **Open EEPROM File...**, or select an `EEPROM` entry in **Bins** and choose **Load EEPROM**.
2. Confirm the detected MS41 layout. If it is unresolved, enable **Manual override** and select the exact MS41.0, MS41.1, MS41.2, or MS41.3 program layout before editing or writing.
3. Select **View / Edit...**. Raw byte editing is disabled by default. Prefer the named transmission shortcut over arbitrary byte edits.
4. For edited checked records, select **Update Checks for Edited Records**. Existing invalid records that were not edited remain unchanged.
5. Select **Apply Changes** to return the edited image to the EEPROM tab. This updates the displayed target only; it does not write to hardware.
6. Use **Save Copy...** when an additional offline copy is required.

ECU Agent

The ECU Agent requires a connected, recognized MS41.0-MS41.3 program with its matching installed Soft-BSL entry and live identity evidence.

1. Select **Read EEPROM...**. A successful stable 512-byte capture is archived in Bins before an additional external copy is offered.
2. Load or edit the exact target and verify that its selected layout matches the connected program.
3. Select **Write Loaded Image...**, review the source and layout, then type `WRITE EEPROM`.
4. The writer saves an immutable before-image, compares before writing, writes only changed bytes with checked-record checks last, and requires an exact full-device readback. If the EEPROM already matches, no write is sent.

If a write becomes uncertain, keep ignition ON and do not disconnect the adapter. Select **Resolve Pending Write...**. The retained session first queries the current state; if remaining prepared operations are required, it asks for `RESUME EEPROM`. After resolution, perform and save a fresh full read before another change.

CH341A Programmer

Disconnect the normal ECU session and keep the ECU unpowered and isolated.

1. Select **Detect CH341A**. Detection enumerates USB devices; it does not read EEPROM.
2. Select **Read EEPROM...** to obtain and archive a stable full-device capture.
3. Load or edit the intended target and confirm its exact layout.
4. Select **Write Loaded Image...** and type `WRITE EEPROM`. The same programmer session performs a stable pre-read, saves the before-image, writes only changed bytes, and verifies the complete target.
5. **Seed ECU Recovery...** is available only for the displayed verified capture when its marker is eligible. **Restore Pre-Seed State...** uses only the exact original bytes captured by that seed operation; it never guesses them.

After reconnecting the programmer or restarting the app, detect and read the EEPROM again. Restore becomes available only when the exact saved original is found; missing or conflicting backups stay blocked.

After an uncertain CH341A write result, perform and save a fresh full read before any further mutation.

Offline files, conversion, and Bins

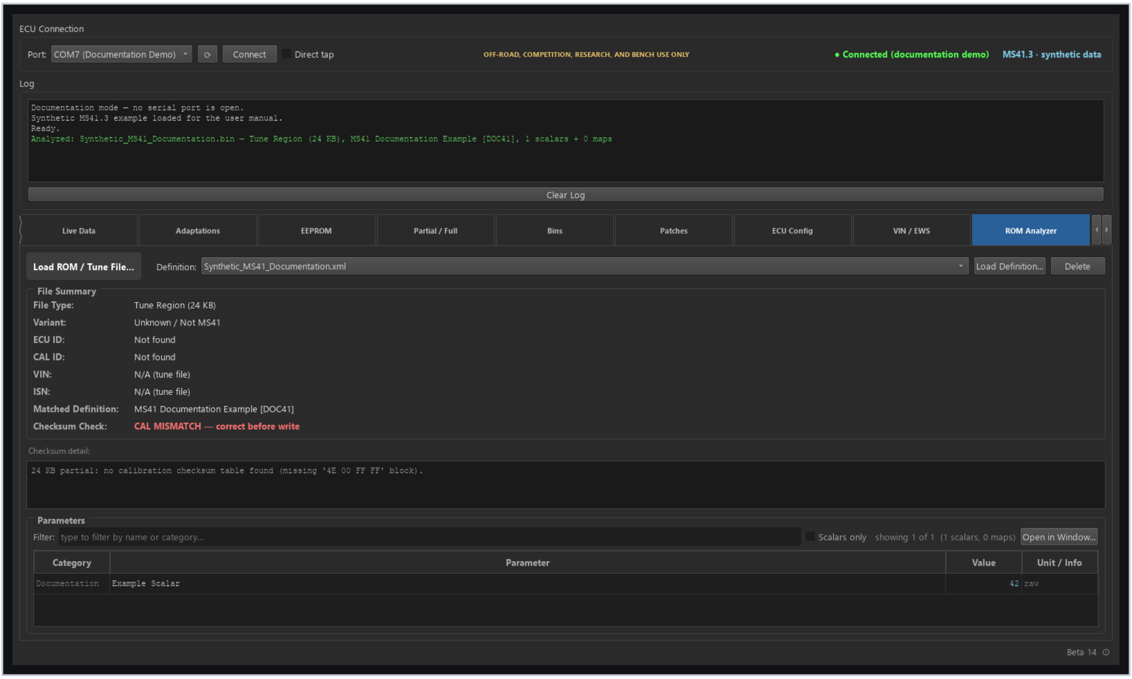
ROM Analyzer

Load a 24 KB or 256 KB BIN to inspect size, version evidence, ECU/CAL identity, VIN, checksum state, and matching definition information. No ECU connection is required.

To enable parameter matching:

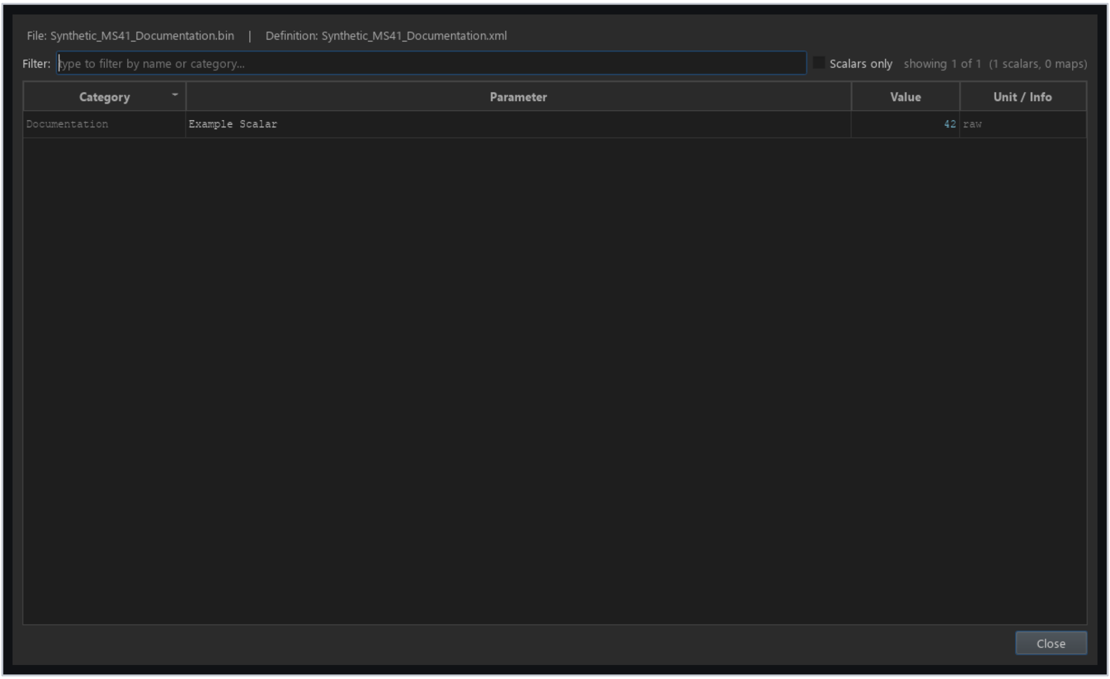
1. Select **Load Definition...** and choose a RomRaider-format MS41 XML definition.
2. The tool validates the XML and copies it into the per-user definition registry.
3. Use the **Definition** list to switch between registered definitions. The selection persists across application restarts.
4. Select **Delete** to remove only the registered copy. The original XML is never modified.

If an imported filename already exists with identical content, the existing copy is selected. If the content differs, the tool asks before replacing the registered copy. Flashing and dumping do not depend on ROM Analyzer definitions.



ROM Analyzer with a registered synthetic definition

Select **Open in Window...** in the Parameters section for a larger, resizable table. The detached window remains modeless, has its own filter and scalar-only control, supports column sorting, and updates automatically when the loaded BIN or selected definition changes.



Detached ROM Analyzer parameters window

Stop if the analyzer reports a hybrid program/calibration combination. A hybrid image can contain code and calibration from different MS41 variants and is blocked from normal flashing.

Partial / Full

- **Full to Partial** extracts the standard 24 KB tune from a 256 KB image.
- **Partial into Full** replaces the calibration in a selected full base.
- Variant conversions preserve or replace identity only according to the explicit workflow. The same conversion policy applies to MS41.0, MS41.1, MS41.2, and MS41.3.

Bins

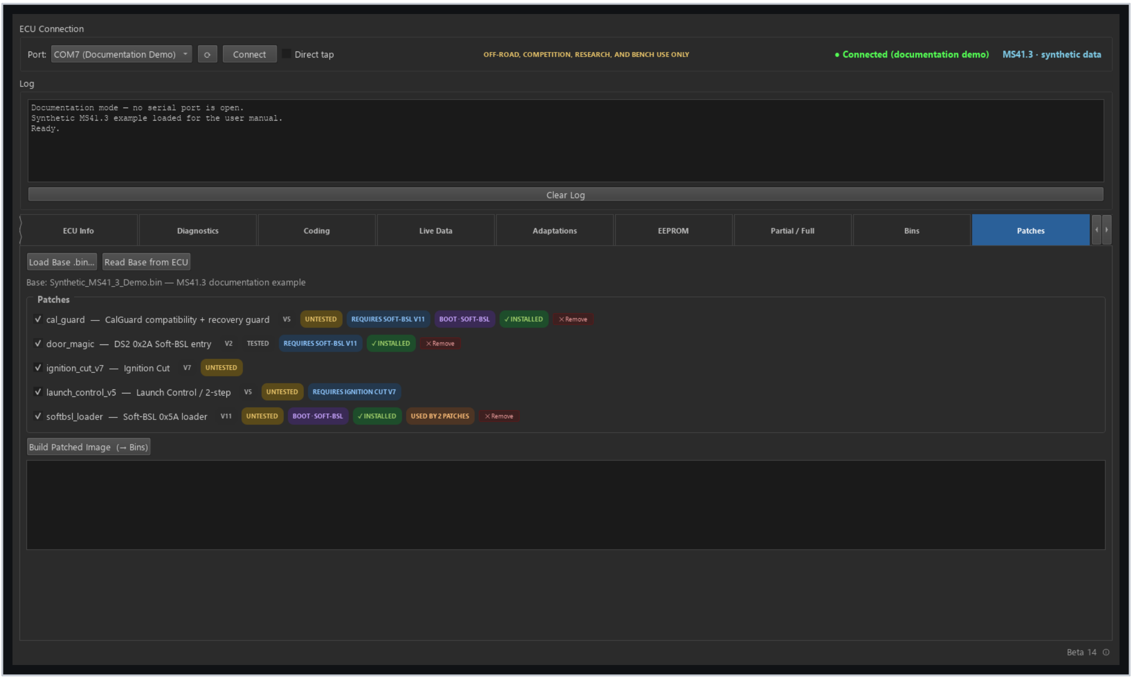
Bins catalogs files created by reads, backups, and patch composition. Entries include available variant, type, VIN/CAL metadata, notes, and source. Use descriptive notes and preserve a known-good original separately from edited or patched images. Newly cataloged files also record a SHA-256 identity so an externally replaced or modified file can be identified later.

Select exactly two entries and choose **Compare** for a read-only report of their SHA-256 identity, program/calibration variants, checksum state, installed patches, changed-byte count, and changed address ranges. A 256 KB full ROM can be compared with a 24 KB tune by comparing only the full ROM's correctly mapped tune region. Legacy entries whose original SHA-256 was never recorded are reported as such; Compare never updates that baseline or changes either file.

Select a 256 KB full-ROM entry and choose **Patches** to load it directly into the Patches tab. Adding or removing patches works on an in-memory copy; **Build Patched Image** archives a new Bin and never overwrites the selected original.

Select a 24 KB tune or 256 KB full-ROM entry and choose **BSL-Unbricker** to load it as the hardware-BSL reference image and open that tab. This prepares the recovery controls only; it does not connect to hardware, review or approve a flash plan, or flash the ECU. A tune selects the **tune** region when available. A full ROM leaves the chip, physical half, and region unchanged for the operator to select explicitly.

Firmware patches



Patch selection with untested status

The Patches tab loads a compatible base image, checks which patches apply, detects already-installed and deprecated revisions, validates dependencies and byte collisions, recomputes required checksums, and archives the composed image into Bins.

Status and migration

- **Installed** means the patch signature is present in the loaded image.
- **Deprecated - remove only** identifies a historical revision retained for safe detection and removal. Deprecated descriptors are not offered for a new installation.
- **Untested** means physical vehicle testing has not been completed.
- Boot-region patches require a transfer path that can actually deliver their bytes.

Ignition Cut V7, Launch Control V4/V5, and AlphaN MAF-failsafe V3 intentionally remain marked **Untested**. Launch Control V4 fuel mode held its configured 4000 RPM setpoint during vehicle testing before the MS41.3 calibration relocation. V5 keeps that native fuel-limiter path, but its relocated MS41.3 controls still require vehicle retesting. Historical variant-specific releases remain visible only when installed so they can be removed before the current Beta 13 revision is applied. Applying one requires an explicit confirmation. Do not treat offline validation as proof of safe behavior on an engine.

> **DANGER: IGNITION CUT HAZARD.** Ignition Cut remains experimental. It may suppress spark while injection continues. Unburned fuel can damage catalytic converters and exhaust components; never use it on a car with catalytic converters.

Bundled patch definition

The release includes `BimmerStein MS41 Patch Definitions.xml` beside the executable. Load it into RomRaider or BimmerStein Tuning Suite to configure calibration items added by the matching patches. Install the firmware patch first and verify the ECU variant, calibration ID, and patch revision. A mismatched definition can expose incorrect tables or write to the wrong calibration addresses. Ignition Cut provides switch and RPM controls. Launch Control adds its mode, clutch, speed, throttle, and limiter settings; fuel mode continues to use the native staged fuel limiter. MS41.3 Launch Control V5 uses its dedicated `0x47E0-0x47E7` block and can be configured with boost control. Deprecated MS41.3 V4 overlapped boost knock-compensation cells; remove it before installing and configuring V5.

Safe composition

1. Load or read a compatible base image.
2. Remove any detected predecessor when instructed.
3. Select the required current patches and dependencies.
4. Review boot-region and Untested badges.
5. Build the image and inspect the build log.
6. Flash the archived result through the normal Flash workflow only after reviewing it.

ECU Config and VIN / EWS

ECU Config

ECU Config exposes supported calibration configuration bits in either file mode or connected-ECU mode. Read the current values first, change only understood options, and write through the normal guarded path. Some options affect checksum verification or core engine features; their warnings are not interchangeable with the optional host read-back Verify checkbox.

Oxygen-feedback choices are resolved from the exact CAL ID because ID41, ID42, ID59, and ID85 do not share one Byte 6 encoding. The ID12/ID60 O2 disable remains experimental and is available only for a full-ROM target. Select **O2 Feedback Program Gate = Feedback Disabled** first; the calibration section will then expose **Oxygen Sensors = Disabled (Experimental)**. Switching the program gate back to enabled removes that calibration choice.

In connected-ECU mode, **Read from ECU** enables the supported program controls. If the current connection already has an archived full read, the application reuses it. Otherwise, a program change makes **Write to ECU** read and archive the ECU's unmodified 256 KB ROM before applying the selected configuration. Calibration-only changes retain the faster 24 KB partial-write route. Program changes are passed to the same guarded full-ROM writer used by the Flash tab, including automatic Soft-BSL/native-fast/standard-DS2 selection and fallback, checksum and flash-family checks, boot-region policy, confirmation, and optional read-back verification. The configuration workflow never uses a different-variant base image as a firmware conversion.

VIN editing

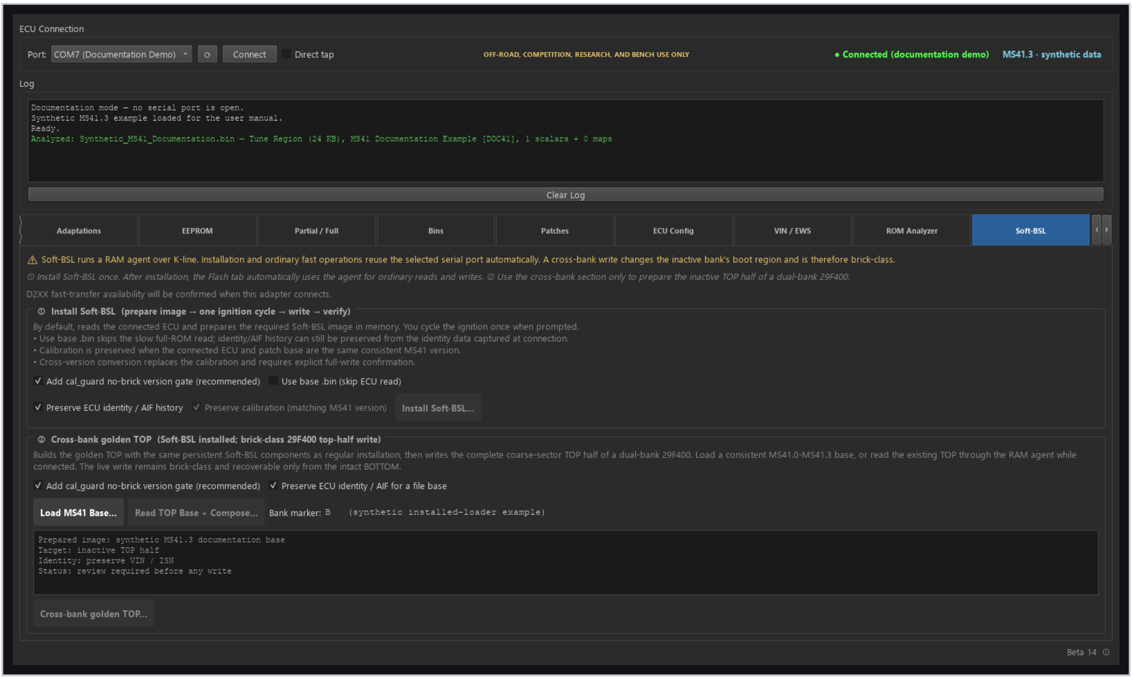
VIN editing is a Soft-BSL sector read-modify-write workflow. It requires a compatible installed loader, detected flash family, D2XX connection, and a live identity-sector cache tied to the current connection. The application changes only the packed VIN field and preserves the rest of the owning erase sector.

EWS2 alignment

EWS alignment is separate from VIN editing. It reads a fresh live DME ISN, applies the validated EWS2 encoding, rechecks the value immediately before transmission, and requires the expected EWS acknowledgement.

> **WARNING:** Do not assume that a cached VIN/identity read proves the current EWS state. VIN and EWS use separate live workflows and separate ownership checks.

Soft-BSL



Soft-BSL installation and recovery controls

Soft-BSL installs a small persistent loader in ECU flash. The loader accepts an agent over DS2, loads it into RAM, and transfers control to that agent for higher-speed reads and writes.

Installation

Use the guided installer. It identifies the target, prepares the temporary entry path, validates the current image and flash family, installs the persistent loader, and confirms normal DS2 communication after the required ignition sequence.

If the temporary Phase 1 write cannot observe the ECU-owned `E659=0xCC` readiness marker, the application closes and releases the adapter before asking for an ignition cycle and an explicit **Retry** or **Cancel**. Cancellation at this prompt is pre-erase: no challenge, selector, erase, or flash command was sent. Each repeated marker timeout requires a new decision; the retry reopens a fresh native-fast transport and never silently changes to the legacy writer.

The temporary installation entry is removed after the persistent loader is written. Subsequent daily operations use the persistent loader and current RAM agents.

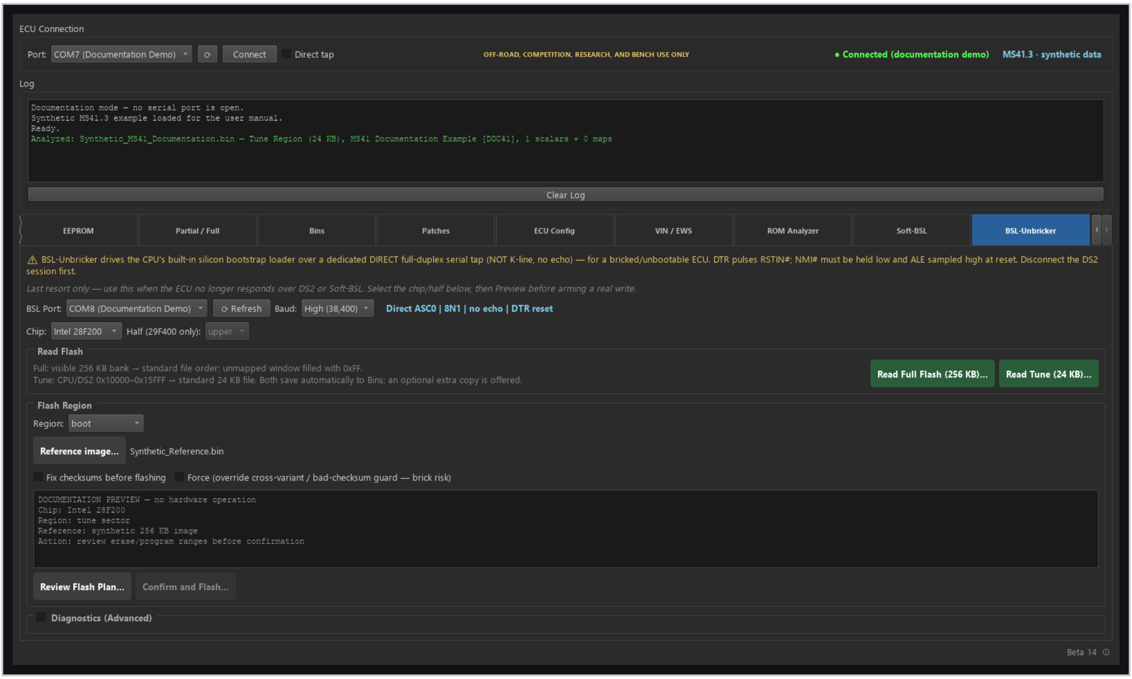
Normal operation

- Full and tune reads use the same optimized RAM-agent path.
- Intel and AMD/JEDEC devices use their matching agent and command set.
- Program writes on MS41.2 include the checksum storage required by its enabled program CRC.
- Successful writes finalize the marker independently of optional host read-back verification.

Cross-bank and boot writes

Golden-bank and boot-region workflows are advanced, recovery-sensitive operations. Follow the displayed A17/bank instructions and verify the selected physical half. Do not use them as a routine replacement for normal tune or program writes.

BSL-Unbricker



Hardware BSL review and programming controls

Hardware BSL communicates through the 80C166 bootstrap loader even when flash contents are blank or corrupt.

Required connection

- Separate full-duplex ASC0 Tx/Rx direct tap.
- 8N1 serial framing with no K-Line echo removal.
- DTR-controlled reset pulse when wired to the supported reset circuit.
- Correct ALE/NMI bootstrap-entry straps.
- Correct chip selection and 29F400 half.

Selectable rates are 9,600, 19,200, and 38,400 baud. Use 38,400 as the preferred D2XX rate and select 19,200 or 9,600 when adapter, wiring, or signal stability requires a fallback.

Intel VPP

Intel 28F200 erase/program requires the correct 12 V VPP/RP# supply. AMD/JEDEC parts do not. The VPP control remains disabled until the Intel chip family is selected.

Read and write workflow

1. Select the dedicated BSL COM port, chip, physical half, region, and rate. The application uses the DTR reset line for the supported entry circuit.
2. For a read, choose full 256 KB file order or the standard 24 KB tune.
3. For a write, select the reference image directly or prepare one from **Bins**, then choose **Review Flash Plan**.
4. Verify every physical address range and prerequisite in the preview.
5. Choose **Confirm and Flash** only when the plan is correct.
6. Monitor erase, programming, and complete physical-region read-back.
7. Remove VPP and bootstrap straps before returning to normal K-Line operation.

Troubleshooting

The COM port is unavailable

- Close other software using the adapter.
- Refresh the port list.
- Confirm whether the normal K-Line adapter or the separate BSL adapter is required.
- Reconnect the FTDI device and verify its Windows driver.

D2XX is unavailable

- Confirm the selected COM port belongs to the intended FTDI device.
- Verify the FTDI D2XX driver is installed.
- Normal DS2 can use the pyserial fallback at 9600.
- Hardware BSL can use its compatible serial fallback. Native-fast DS2 requires D2XX; Soft-BSL can fall back to its supported low-rate serial tier when D2XX is unavailable.

The fast-path stability check fails

If a Soft-BSL tier fails before erase, the application can retry a lower Soft-BSL tier. If a native-fast DS2 check fails before erase and normal ECU state is confirmed, it can restart the complete operation over normal DS2. Check adapter latency, wiring, ground, ECU voltage, and signal integrity before retrying.

A write failed

- If the UI says erase did not start, correct the reported connection or file problem and retry.
- If the UI says recovery is active, keep ignition ON and use **Retry Flash Recovery**.
- If the ECU no longer boots and no retained session exists, prepare the hardware-BSL connection.

The ROM Analyzer cannot load definitions

Use **Load Definition...** to select a valid RomRaider-format MS41 XML file. Do not copy XML files into `_internal` or any other packaged runtime directory. If a registered definition was changed or damaged outside the application, delete it and import a known-good copy again. Confirm the BIN size and exact ECU software identity before relying on matched values.

Values or identity look wrong

Stop the operation. Confirm the connected ECU, file, variant, byte order, and definition. Re-read the value through an independent path before writing anything.

Exporting a support/test bundle

Choose the build label at the lower-right of the main window, then **Export Support Bundle**. The ZIP contains build information plus, when available, privacy-scoped metadata and SHA-256 for the single selected Bin, the latest Live Data CSV, and the latest native-fast journal. Raw ROM bytes are never included.

A session log is optional because it may contain VIN, ISN, ECU identifiers, local paths, and detailed errors. The application asks before including it. Bin filenames, VIN, ISN, notes, and ECU identifiers are excluded from the default metadata, and generic ZIP member names are used.

Final checklists

Before a normal read

[ ] ECU voltage is stable.

[ ] The normal K-Line/direct-tap selection matches the wiring.

[ ] The correct COM port is selected.

[ ] Full 256 KB or tune 24 KB is intentionally selected.

[ ] The output location does not overwrite an irreplaceable file.

Before a normal write

[ ] The target file size and ECU variant match the intended job.

[ ] The detected flash-chip family is compatible with the image and path.

[ ] Checksum correction is enabled unless there is a documented reason otherwise.

[ ] Backup and Verify choices reflect the operator's decision.

[ ] Stable power and hardware recovery are available.

[ ] No other software owns the COM port.

Before hardware BSL programming

[ ] Direct ASC0 wiring and bootstrap straps are correct.

[ ] Chip family and 29F400 half are correct.

[ ] Intel VPP is active only for the Intel write phase.

[ ] The reviewed physical address plan matches the intended region.

[ ] The reference image is the correct size and byte order.

[ ] A recovery plan exists if programming is interrupted.

Disclaimer

BimmerStein ECU Tool is experimental software intended solely for off-road, competition, research, and bench use. It is provided "as is," without warranty of any kind, to the maximum extent permitted by applicable law.

ECU programming, calibration changes, firmware patches, and recovery operations can cause data loss, an unbootable ECU, engine or vehicle damage, unexpected engine behavior, or unsafe operating conditions. The user assumes all risks associated with connecting, configuring, modifying, or flashing an ECU and is responsible for maintaining suitable backups, stable power, and an appropriate recovery method.

The user is solely responsible for determining whether any operation or modification is legal and compliant with applicable emissions, safety, registration, competition, and other regulations. An off-road designation does not establish that a particular modification is lawful.

BimmerStein ECU Tool is independent software and is not affiliated with or endorsed by BMW AG, FTDI, or RomRaider. Nothing in this disclaimer limits the rights granted under the GNU General Public License version 3.

Project and license

BimmerStein ECU Tool is independent software for compatible BMW MS41 workflows. It is not affiliated with or endorsed by BMW AG, FTDI, or RomRaider.

The software is intended solely for off-road, competition, research, and bench use. It is not designed or certified for modifying a vehicle operated on public roads.